DOI 10.1007/s10986-018-9414-3
Lithuanian Mathematical Journal, Vol. 58, No. 4, October, 2018, pp. 360–378

Computation of p-variation*

Vygantas Butkus and Rimas Norvaiša

Faculty of Mathematics and Informatics, Vilnius University, Naugarduko str. 24, LT-08663 Vilnius, Lithuania
(e-mail: vygantas.butkus@gmail.com; rimas.norvaisa@mii.vu.lt)

Received May 28, 2018; revised June 27, 2018

Abstract. A code for computing the p-variation of a piecewise monotone function is introduced. The code is publicly available in the R environment package under the name *pvar*. The algorithm is based on some properties of the p-variation of a piecewise monotone function proved in this paper. The mathematical results may have their own interest.

MSC: 26A45

Keywords: p-variation, computation, piecewise monotone function

1 Introduction

Let f be a real-valued function defined on an interval $[0,1]$, and let $0<p<\infty$. Given a partition $\kappa=(x_i)_{i=0}^n$ of $[0,1]$, the p-variation sum $\sum_{i=1}^n |f(x_i)-f(x_{i-1})|^p$ is denoted by $s_p(f,\kappa)$. The p-variation of f is the quantity

$$v_p(f) := \sup\bigl\{s_p(f,\kappa)\colon \kappa \text{ is a partition of } [0,1]\bigr\}, \tag{1.1}$$

which may be finite or infinite. The 1-variation is the total variation of f. If the total variation of f is infinite, then it may have bounded p-variation for some $p>1$. In this case, the function f is called a rough function or a rough path.

The 2-variation was introduced and used in the theory of Fourier series by N. Wiener (see [15]). p-variation and more generally Φ-variation were introduced and used by L.C. Young to extend the Stieltjes-type integration (see, e.g., [16]). During last twenty years or so, a theory of integral equations have been extended to the class of rough functions (see e.g. [6, 7, 8]).

A sample path of Wiener process $W=\{W(t)\colon 0\leqslant t\leqslant 1\}$ is a typical example of a rough function. If $p>2$, then almost all sample paths of W have bounded p-variation. By the results of Norvaiša and Račkauskas [10], $v_p(W)$ is a limit in distribution of a sequence of random variables $(v_p(S_n))$, where S_n is a partial-sum process with values $S_n(t) := X_1+\cdots+X_{\lfloor t\rfloor}$, $0\leqslant t\leqslant 1$, and (X_i) is a sequence of independent identically distributed random variables with mean zero and finite second moment. This fact suggests a possibility to evaluate the p-variation of a sample function of Wiener process using the values of p-variation of sample functions of a partial-sum process. This was our motivation to find an algorithm that could allow us to calculate the p-variation of a piecewise constant function as fast as possible.

* The research supported by the Research Council of Lithuania, grant No. S-MIP-17-76.

To compute the p-variation of a function f, we need to know first that the supremum in (1.1) is achievable, and if so, then we need to find a maximizing partition κ. If f is a piecewise monotone function with all points of its local extrema making a partition $\kappa = (x_i)$ and if $p = 1$, then the supremum in (1.1) is achieved by the partition κ. If $p > 1$, then finding a maximizing partition is not an obvious task as the following example shows. Let f be the function with values

$$f(x) = 0\mathbf{1}_{[0,\,1/3)}(x) + \frac{2}{3}\mathbf{1}_{[1/3,\,2/3)}(x) + \frac{1}{3}\mathbf{1}_{[2/3,\,1)}(x) + \mathbf{1}_{\{1\}}(x)$$

for $x \in [0, 1]$, where $\mathbf{1}_A$ is the indicator function of a set A. Taking $p = 3$, $\kappa_1 = (0, 1/3, 2/3, 1)$, and $\kappa_2 = (0, 1)$, we have the relation

$$s_3(f, \kappa_1) = \frac{17}{27} < 1 = s_3(f, \kappa_2) = v_3(f)$$

for the 3-variation sums. Therefore a maximizing partition need not be formed by all points of local extrema when $p > 1$.

The main result of this paper is an algorithm that searches for a maximizing partition of the Φ-variation of a function. The algorithm is already "realized" in the R environment (see [14]) and publicly available in the *pvar* package on CRAN (see [1]). The next section provides some mathematical properties of partitions that maximize the Φ-variation of a piecewise monotone function. The algorithm is constructed in Section 3.

2 Φ-variation of piecewise monotone functions

In this section, we prove the main properties of partitions which maximize Φ-variation of a piecewise monotone function. The main result of this section is Theorem 3. It helps to identify points of local extrema of a function that do not belong to a maximizing partition. The results of this section are used in Section 3 for a piecewise constant function and Φ being a power function.

2.1 Notation and auxiliary results

Regulated functions. Let $[a, b]$ be a closed interval of real numbers, and let $a < b$. A real-valued function f defined on $[a, b]$ is *regulated* if the left limit $f(x-) := \lim_{y\uparrow x} f(y)$ exists for each $x \in (a, b]$ and the right limit $f(x+) := \lim_{y\downarrow x} f(y)$ exists for each $x \in [a, b)$. We extend regulated functions to a subset of the set lexicographically ordered $\mathbb{R}^{\pm} := \mathbb{R}\times\{-1, 0, +1\}$. Given $x \in \mathbb{R}$, the elements $(x, -1) \in \mathbb{R}^{\pm}$ and $(x, +1) \in \mathbb{R}^{\pm}$ are denoted by $x-$ and $x+$, respectively, whereas $(x, 0)$ is identified with x. Via this identification, $\mathbb{R}$ and all its subsets are also subsets of $\mathbb{R}^{\pm}$. Given $a \in \mathbb{R}^{\pm}$ and $b \in \mathbb{R}^{\pm}$ such that $a \leqslant b$, let $[a, b]^{\pm} := \{x \in \mathbb{R}^{\pm}\colon a \leqslant x \leqslant b\}$. If $a \in \mathbb{R}$ and $b \in \mathbb{R}$, then the set $[a, b]^{\pm}$ consists of points a, $a+$, $x-$, x, $x+$ for $x \in (a, b)$ and of $b-$, b with the natural linear ordering: $x- < x < x+ < y-$ if $x < y$. The set $[a, b]^{\pm}$ is obtained by augmenting $[a, b]$, and so we call $[a, b]^{\pm}$ the *augmentation of* $[a, b]$. Each regulated function f on $[a, b]$ extends naturally to the augmented interval $[a, b]^{\pm}$. Throughout the paper, we do not distinguish f from its extension on $[a, b]^{\pm}$.

The augmented interval $[a, b]^{\pm}$ is equipped with the order topology. The neighborhoods of each point in $[a, b]^{\pm}$ are of the form $U(x-) = [(x - 1/n)+, x-]^{\pm}$ with $n \in \mathbb{N}$, $U(x+) = [x+, (x + 1/n)-]^{\pm}$ with $n \in \mathbb{N}$, and the points $x = (x, 0)$ are isolated, $U(x) = \{x\}$. The set $[a, b]^{\pm}$ is compact and has $[a, b]$ as a dense open discrete subset. The set $\mathcal{R}[a, b]$ of all regulated functions on $[a, b]$ can be identified with (as an isometrical isomorphic copy of) the set of all continuous real-valued functions on $[a, b]^{\pm}$ (see [2]).

In this paper, we consider piecewise monotone functions that are regulated. For example, if f on $[a, b]$ is nondecreasing and $x \in (a, b)$, then

$$f(x-) = \sup\bigl\{f(y)\colon a \leqslant y < x\bigr\} \leqslant f(x) \leqslant f(x+) = \inf\bigl\{f(y)\colon x < y \leqslant b\bigr\}.$$

Similar inequalities hold for nonincreasing functions.

Partitions of intervals. Let J be either a closed interval $[a, b]$ with $a < b$ or its augmentation $[a, b]^{\pm}$. A tuple $\kappa = (x_i)_{i=0}^n = (x_0, x_1, \dots, x_n)$ of elements of J is called a *partition of* J if $a = x_0 < x_1 < \dots < x_n = b$. Each x_i is called a *partition point* of κ, and each $[x_{i-1}, x_i] = \{x \in J\colon x_{i-1} \leqslant x \leqslant x_i\}$ is called a *partition interval* of κ. If a partition point x_i is not an endpoint a or b, then it is called an *inner partition point*. The number of partition intervals is called the *size* of the partition κ and is denoted by $|\kappa|$. Thus $|\kappa|$ is one less than the cardinality of the set partition points $\{x_i\colon 0 \leqslant i \leqslant n\}$. Let λ and κ be two partitions of J. If each partition point of λ is a partition point of κ, then we say that κ is a *refinement* of λ, or that λ is a *subpartition* of κ, and write $\lambda \sqsubset \kappa$. If $\lambda = (x_i)_{i=0}^n$ is a partition of J_1, $\kappa = (y_i)_{i=0}^m$ is a partition of J_2, and the right endpoint of J_1 is the left endpoint of J_2, then the partition $\lambda \sqcup \kappa := (x_0, x_1, \dots, x_n (= y_0), y_1, \dots, y_m)$ of $J_1 \cup J_2$ is a *merger* of λ and κ. The set of all partitions of J is denoted by $\mathcal{P}(J)$.

Φ-variation. Let $\mathcal{V}$ be the class of all functions $\Phi\colon [0, \infty) \to [0, \infty)$ that are strictly increasing, continuous, unbounded, and 0 at 0. Let $\mathcal{CV}$ be the subclass of convex functions in $\mathcal{V}$. Let f be a real-valued function defined on an interval $[a, b]$, and let $\Phi \in \mathcal{V}$. If $a < b$, then, for a partition $\kappa = (x_i)_{i=0}^n$ of $[a, b]$, the *Φ-variation sum* is

$$s_\Phi(f, \kappa) := \sum_{i=1}^n \Phi\bigl(\bigl|f(x_i) - f(x_{i-1})\bigr|\bigr). \tag{2.1}$$

The *Φ-variation* of f over $[a, b]$ is 0 if $a = b$, and otherwise

$$v_\Phi(f) = v_\Phi\bigl(f, [a, b]\bigr) := \sup\bigl\{s_\Phi(f, \kappa)\colon \kappa \in \mathcal{P}[a, b]\bigr\}. \tag{2.2}$$

We say that f has bounded Φ-variation if $v_\Phi(f) < \infty$. In the case $\Phi(u) = u^p$, $u \geqslant 0$, for some $0 < p < \infty$, (2.1) is the p-variation sum defined earlier, (2.2) is the p-variation of f and agrees with (1.1) if in addition $[a, b] = [0, 1]$.

By Proposition 3.33 in [6] each function f with bounded Φ-variation is regulated. Therefore it makes sense to define $v_\Phi(f, [a, b]^{\pm})$ as in (2.2) except that the class of partitions $\mathcal{P}[a, b]$ is replaced by the class $\mathcal{P}([a, b]^{\pm})$ of partitions of the augmented interval $[a, b]^{\pm}$. Using arguments to prove (2.13) in [5, p. 84], it follows that

$$v_\Phi\bigl(f, [a, b]\bigr) = v_\Phi\bigl(f, [a, b]^{\pm}\bigr),$$

provided that either side is finite.

DEFINITION 1. Let $\Phi \in \mathcal{V}$, and let $f : [a, b] \to \mathbb{R}$ be regulated. A partition κ of $[a, b]^{\pm}$ is called a *maximizing partition* for the Φ-variation of f if $v_\Phi(f, [a, b]) = s_\Phi(f, \kappa)$. The set of all such partitions is denoted by $\mathcal{MP}_\Phi(f, [a, b])$.

Piecewise monotone functions. Let f be a real-valued function defined on $[a, b]$ with endpoints $a < b$, and let $\kappa = (t_i)_{i=0}^n$ be a partition of $[a, b]$. We say that f is *piecewise monotone* (*piecewise constant*) *with respect to* κ if

$$f \text{ is monotone (constant) on each open interval } (t_{i-1}, t_i),\ i \in \{1, \dots, n\}. \tag{2.3}$$

Also, we say that f is *piecewise monotone* (*piecewise constant*) if it is piecewise monotone (piecewise constant) with respect to κ for some $\kappa \in \mathcal{P}([a, b])$. The set of all piecewise monotone functions on $[a, b]$ is denoted by $\mathcal{PM}[a, b]$. Clearly, each piecewise constant function is a piecewise monotone function. Also, each piecewise monotone function is regulated, and so it has the extension to the augmentation of its interval of definition.

Given $\kappa = (t_i)_{i=0}^n \in \mathcal{P}([a, b])$, $\kappa^{\pm} := (t_0, t_0+, t_1-, t_1, t_1+, \dots, t_n-, t_n)$ is a partition of $[a, b]^{\pm}$. On the partition intervals $[t-, t]$ and $[t, t+]$, each regulated function assumes at most two values, and so it is monotone when restricted to such an interval. Let f be a regulated function on $[a, b]$. Clearly, f is piecewise monotone if and only if there exists $\lambda \in \mathcal{P}([a, b]^{\pm})$ such that

$$f \text{ is monotone on each partition interval of } \lambda. \tag{2.4}$$

If (2.4) holds, then we say that λ is a *monotonicity partition for f* and f is a piecewise monotone function with respect to λ.

If a regulated function f on $[a,b]$ is piecewise monotone, then there is a partition of $[a,b]^{\pm}$ with minimal size for which (2.4) holds. This follows from the fact that each nonempty set of positive integers has a minimal element. We denote the minimal size by $K(f)$, and so

$$K(f) = \min\bigl\{|\lambda|\colon \text{(2.4) holds for some } \lambda \in \mathcal{P}\bigl([a,b]^{\pm}\bigr)\bigr\}. \tag{2.5}$$

We say that $\lambda \in \mathcal{P}([a,b]^{\pm})$ is a *minimal monotonicity partition for f* if (2.4) holds and $|\lambda| = K(f)$. The set of all such λ is denoted by $\varXi(f)$.

For example, let $f : [0,1] \to \mathbb{R}$ be a function with values $f(x) = x$ for $x \in [0,1)$ and $f(1) = 0$. Then f is piecewise monotone with respect to the partition $\lambda = (0, 1-, 1)$ of $[0,1]^{\pm}$ since it is increasing on the partition interval $[0, 1-]$ and is decreasing on the partition interval $[1-, 1]$. Also, $\lambda \in \varXi(f)$ with $K(f) = 2$.

For each partition in $\varXi(f)$, its every inner partition point is a strict local extremum of f as the following shows.

Proposition 1. (*See* [12, *Prop*. 3.2].) *Let $f \in \mathcal{PM}[a,b]$. If $(t_j)_{j=0}^{K(f)} \in \varXi(f)$, then exactly one of the following statements holds*:

(i) $f(t_0) > f(t_1) < f(t_2) > \cdots$, *$f$ is nonincreasing* (*nondecreasing*) *on $[t_{j-1}, t_j]$ when $j \in \{1, \dots, K(f)\}$ is an odd* (*even*) *integer*; *or*
(ii) *(i) holds for $-f$*; *or*
(iii) $K(f) = 1$*, and f is constant on* $[a,b]$.

The $\varPhi$-variation of a piecewise monotone function f is achievable on a subpartition of any minimal monotonicity partition of f. This fact may be a part of mathematical folklore. Due to its importance for the present paper, we include the proof given by J. Qian in unpublished PhD thesis [12, Thm. 3.1]. A different proof for piecewise linear (and so continuous) functions can be found in the lecture notes [3, p. 11].

Theorem 1. *Let $\varPhi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a,b]$, and let $\lambda = (t_j)_{j=0}^{K(f)} \in \varXi(f)$. Then*

$$v_{\varPhi}\bigl(f, [a,b]\bigr) = \max\bigl\{s_{\varPhi}(f, \kappa)\colon (a,b) \sqsubset \kappa \sqsubset \lambda\bigr\}.$$

Proof. Let $\zeta = (s_i)_{i=0}^{n}$ be any partition of $[a,b]$. It suffices to prove that there is a subpartition κ of the minimal monotonicity partition λ such that the $\varPhi$-variation sum $s_{\varPhi}(f,\zeta) \leqslant s_{\varPhi}(f,\kappa)$. Since $\varPhi$ is convex and $\varPhi(0) = 0$, we have the superadditivity property: $\varPhi(u+v) \geqslant \varPhi(u) + \varPhi(v)$ for any $u, v \geqslant 0$ (cf. [6, p. 104]). Using this property, we can show that without loss of generality we can assume:

(i) there are at most two partition points of ζ in each $[t_{j-1}, t_j]$, $1 \leqslant j \leqslant K(f)$;
(ii) the increments $f(s_i) - f(s_{i-1})$, $1 \leqslant i \leqslant n$, have alternating signs.

Given $j \in \{1, \dots, K(f)\}$, the interval $[t_{j-1}, t_j]$ contains either two points of ζ, or one point of ζ, or no points of ζ. We show that we can replace the points of ζ by points of λ without decreasing the $\varPhi$-variation sum $s_{\varPhi}(f,\zeta)$.

First, suppose that there are two partition points s_{i_1} and s_{i_2} of ζ such that $t_{j-1} \leqslant s_{i_1} < s_{i_2} \leqslant t_j$. Then

$$\begin{aligned}
\bigl|f(s_{i_2}) - f(s_{i_1})\bigr| &\leqslant \bigl|f(t_j) - f(t_{j-1})\bigr|,\\
\bigl|f(s_{i_1}) - f(s_{i_1-1})\bigr| &\leqslant \bigl|f(t_{j-1}) - f(s_{i_1-1})\bigr|,\\
\bigl|f(s_{i_2+1}) - f(s_{i_2})\bigr| &\leqslant \bigl|f(s_{i_2+1}) - f(t_j)\bigr|.
\end{aligned}$$

Thus, replacing s_{i_1} by t_{j-1} and s_{j_2} by t_j, we obtain a new partition ζ' of $[a,b]$ such that

$$s_{\varPhi}(f,\zeta) \leqslant s_{\varPhi}(f,\zeta'). \tag{2.6}$$

Second, suppose that $[t_{j-1}, t_j]$ contains just one partition point s_i of ζ. Now we have four subcases. First, if $f(s_i) > f(s_{i-1})$ and $f(t_j) > f(t_{j-1})$, then $f(s_{i+1}) < f(s_i)$ by (ii), and so

$$\begin{aligned}\left|f(s_i) - f(s_{i-1})\right| &\leqslant \left|f(t_j) - f(s_{i-1})\right|,\\ \left|f(s_{i+1}) - f(s_i)\right| &\leqslant \left|f(s_{i+1}) - f(t_j)\right|.\end{aligned}$$

Thus, replacing s_i by t_j, we obtain a new partition ζ' of $[a, b]$ such that (2.6) holds. Similar arguments yield the same bound (2.6) in the other three subcases. Making these changes for each j, we obtain a subpartition κ of λ such that $s_\Phi(f, \zeta) \leqslant s_\Phi(f, \kappa)$. The proof of Theorem 1 is complete. □

By Theorem 1 a maximizing partition is a subpartition of a given minimal monotonicity partition. Since a piecewise monotone function has at least one minimal monotonicity partition, a maximizing partition always exists as stated next.

Corollary 1. *For $\Phi \in \mathcal{CV}$ and any $f \in \mathcal{PM}[a, b]$, the set of maximizing partitions $\mathcal{MP}_\Phi(f, [a, b])$ is nonempty.*

It is known that the only continuous functions that have bounded p-variation for some $p \in (0, 1)$ are constants. Therefore if $\Phi \in \mathcal{V}$ is concave, then we consider the Φ-variation only for piecewise constant functions.

Theorem 2. *Let $\Phi \in \mathcal{V}$ be concave, and let $f : [a, b] \to \mathbb{R}$ be piecewise constant with respect to a partition $\kappa = (t_i)_{i=0}^n$ of $[a, b]$. Let $r_{2i-1} \in [t_{i-1}+, t_i-]$ for $i \in \{1, \dots, n\}$, $r_{2i} := t_i$ for $i \in \{0, 1, \dots, n\}$, and $\lambda := (r_j)_{j=0}^{2n} \in \mathcal{P}([a, b]^\pm)$. Then $v_\Phi(f) = s_\Phi(f, \lambda)$. If, in addition, f is right-continuous, then $v_\Phi(f) = s_\Phi(f, \kappa)$.*

Proof. Let $\zeta = (s_k)_{k=0}^l$ be any partition of $[a, b]$. To prove the first part, it suffices to show that

$$s_\Phi(f, \zeta) \leqslant s_\Phi(f, \lambda). \tag{2.7}$$

Each s_k in $s_\Phi(f, \zeta)$ can be replaced by some partition point r_j from λ without changing the value of f. Let λ' be the partition of $[a, b]^\pm$ obtained by all different such r_j. If $\lambda' = \lambda$, then (2.7) holds. Otherwise, $\lambda' \sqsubset \lambda$, and there is a partition point r of λ that is not a partition point of λ'. Let λ'' be λ' with r included as a partition point. Since Φ is concave and $\Phi(0) = 0$, we have the subadditivity property: $\Phi(u + v) \leqslant \Phi(u) + \Phi(v)$ for any $u, v \geqslant 0$. Therefore $s_\Phi(f, \zeta) = s_\Phi(f, \lambda') \leqslant s_\Phi(f, \lambda'')$. If $\lambda'' = \lambda$, then (2.7) holds. Otherwise, repeating the same argument finitely many times, (2.7) follows.

Now suppose that f is right-continuous. Then $f(r_{2i-2}) = f(t_{i-1}) = f(t_{i-1}+) = f(r_{2i-1})$ for each $i \in \{1, \dots, n\}$. Therefore $s_\Phi(f, \lambda) = s_\Phi(f, \kappa)$, and the second part of the theorem is proved. □

2.2 Partition points of maximizing partitions

If a function $f : [a, b] \to \mathbb{R}$ has bounded Φ-variation, then

$$v_\Phi\bigl(f; [a, b]\bigr) \geqslant \sum_{i=1}^n v_\Phi\bigl(f, [t_{i-1}, t_i]\bigr) \tag{2.8}$$

for every partition $(t_i)_{i=0}^n$ of $[a, b]$. We show that equality (2.8) can be used to describe a subpartition of a maximizing partition.

Proposition 2. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a, b]$ for some $a, b \in \mathbb{R}$ with $a < b$, and let $\lambda = (t_i)_{i=0}^n$ be a partition of $[a, b]^\pm$. Then λ is a subpartition of some $\kappa \in \mathcal{MP}_\Phi(f, [a, b])$ if and only if*

$$v_\Phi\bigl(f, [a, b]\bigr) = \sum_{i=1}^n v_\Phi\bigl(f, [t_{i-1}, t_i]\bigr). \tag{2.9}$$

If, in addition, λ is a subpartition of some minimal monotonicity partition μ of f, then (2.9) *holds if and only if there is $\kappa \in \mathcal{MP}_\Phi(f,[a,b])$ such that $\lambda \sqsubset \kappa \sqsubset \mu$.*

Proof. To prove the first part, suppose $\kappa = (r_j)_{j=0}^m \in \mathcal{MP}_\Phi(f,[a,b])$, $\lambda \sqsubset \kappa$, and $r_{j(i)} = t_i$ for each $i \in \{0,1,\dots,n\}$. Then for each i, $(r_j)_{j=j(i-1)}^{j(i)}$ is a partition of $[t_{i-1},t_i]$, and

$$v_\Phi\bigl(f,[a,b]\bigr) = \sum_{i=1}^{n} \sum_{j=j(i-1)+1}^{j(i)} \Phi\bigl(\bigl|f(r_j)-f(r_{j-1})\bigr|\bigr) \leqslant \sum_{i=1}^{n} v_\Phi\bigl(f,[t_{i-1},t_i]\bigr).$$

This, together with (2.8), gives (2.9).

Now suppose that (2.9) holds. For each $i \in \{1,\dots,n\}$, the restriction f_i of f to $[t_{i-1},t_i]$ is a piecewise monotone function. By Corollary 1 each $\mathcal{MP}_\Phi(f_i,[t_{i-1},t_i])$ is nonempty. Let $\kappa_i \in \mathcal{MP}_\Phi(f_i,[t_{i-1},t_i])$ for each i, and let $\kappa := \sqcup_{i=1}^n \kappa_i$. Then $\lambda \sqsubset \kappa$, and by (2.9) we have

$$v_\Phi\bigl(f,[a,b]\bigr) = \sum_{i=1}^{n} v_\Phi\bigl(f_i,[t_{i-1},t_i]\bigr) = \sum_{i=1}^{n} s_\Phi(f_i,\kappa_i) = s_\Phi(f,\kappa).$$

The proof of the first part is complete. The proof of the second part is similar except that Theorem 1 is used instead of Corollary 1. □

Corollary 2. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a,b]$ for some $a,b \in \mathbb{R}$ with $a < b$, and let $\kappa = (t_i)_{i=0}^n \in \mathcal{MP}_\Phi(f,[a,b])$. For any pair of integers k,l such that $0 \leqslant k < l \leqslant n$,*

$$v_\Phi\bigl(f,[t_k,t_l]\bigr) = \sum_{i=k+1}^{l} v_\Phi\bigl(f,[t_{i-1},t_i]\bigr). \tag{2.10}$$

Proof. Let k,l be a pair of integers such that $0 \leqslant k < l \leqslant n$. First, using the "only if" part of Proposition 2 for the partition $\lambda = (t_0,t_k,t_{k+1},\dots,t_{l-1},t_l,t_n) \sqsubset \kappa$, we have

$$v_\Phi\bigl(f,[a,b]\bigr) = v_\Phi\bigl(f,[t_0,t_k]\bigr) + \sum_{i=k+1}^{l} v_\Phi\bigl(f,[t_{i-1},t_i]\bigr) + v_\Phi\bigl(f,[t_l,t_n]\bigr).$$

Second, using the same implication for the partition $\lambda = (t_0,t_k,t_l,t_n) \sqsubset \kappa$, we have

$$v_\Phi\bigl(f,[a,b]\bigr) = v_\Phi\bigl(f,[t_0,t_k]\bigr) + v_\Phi\bigl(f,[t_k,t_l]\bigr) + v_\Phi\bigl(f,[t_l,t_n]\bigr).$$

Combining the two equalities, (2.10) follows. □

Corollary 3. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a,b]$ for some $a,b \in \mathbb{R}$ with $a < b$, and let $\kappa = (t_i)_{i=0}^n \in \mathcal{MP}_\Phi(f,[a,b])$. For any pair of integers k,l such that $0 \leqslant k < l \leqslant n$,*

$$v_\Phi\bigl(f,[t_k,t_l]\bigr) = s_\Phi\bigl(f,(t_i)_{i=k}^l\bigr). \tag{2.11}$$

Proof. Let k,l be a pair of integers such that $0 \leqslant k < l \leqslant n$. First, since κ is maximizing partition, we have

$$\begin{aligned} v_\Phi\bigl(f,[a,b]\bigr) = s_\Phi\bigl(f,(t_i)_{i=0}^n\bigr) &= s_\Phi\bigl(f,(t_i)_{i=0}^k\bigr) + s_\Phi\bigl(f,(t_i)_{i=k}^l\bigr) + s_\Phi\bigl(f,(t_i)_{i=l}^n\bigr) \\ &\leqslant v_\Phi\bigl(f,[a,t_k]\bigr) + s_\Phi\bigl(f,(t_i)_{i=k}^l\bigr) + v_\Phi\bigl(f,[t_l,b]\bigr). \end{aligned} \tag{2.12}$$

Second, by the "only if" part of Proposition 2 for the partition $\lambda = (t_0, t_k, t_l, t_n) \sqsubset \kappa$ we have

$$v_\Phi\big(f, [a,b]\big) = v_\Phi\big(f, [a, t_k]\big) + v_\Phi\big(f, [t_k, t_l]\big) + v_\Phi\big(f, [t_l, b]\big). \tag{2.13}$$

Combining (2.12) and (2.13), the inequality

$$v_\Phi\big(f, [t_k, t_l]\big) \leqslant s_\Phi\big(f, (t_i)_{i=k}^l\big)$$

follows, which yields (2.11). $\square$

DEFINITION 2. Let $\Phi \in \mathcal{V}$, let $f \in \mathcal{PM}[a,b]$ for some $a, b \in \mathbb{R}$ with $a < b$, and let $x \in [a,b]^\pm$. We say that x is a *maximizing partition point for the Φ-variation of f* if x is a partition point of some maximizing partition for the Φ-variation of f. The set of all such points is denoted by $\mathcal{MPP}_\Phi(f, [a,b])$. If $x \in [a,b]^\pm \setminus \mathcal{MPP}_\Phi(f, [a,b])$, then we say that x is a *redundant point for the Φ-variation of f*, or just a *redundant point*.

For example, given $(t_i)_{i=0}^n \in \mathcal{MP}_\Phi(f, [a,b])$, each t_i is a maximizing partition point for the Φ-variation of f.

Lemma 1. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a,b]$ for some $a, b \in \mathbb{R}$ with $a < b$, let $x \in [a,b]^\pm \setminus \mathcal{MPP}_\Phi(f, [a,b])$, and let $(t_i)_{i=0}^n \in \mathcal{MP}_\Phi(f, [a,b])$. There is $j \in \{1, \ldots, n\}$ such that*

$$t_{j-1} < x < t_j \quad \textit{and} \quad x \notin \mathcal{MPP}_\Phi\big(f, [t_{j-1}, t_j]\big). \tag{2.14}$$

Proof. Since $(t_i)_{i=0}^n$ is a maximizing partition and x is not a maximizing partition point, the first part of (2.14) must hold for some $j \in \{1, \ldots, n\}$. To prove the second part, for the same j, suppose that $x \in \mathcal{MPP}_\Phi(f, [t_{j-1}, t_j])$. Then there is $\kappa \in \mathcal{MP}_\Phi(f, [t_{j-1}, t_j])$ such that x is a partition point of κ and

$$v_\Phi\big(f, [t_{j-1}, t_j]\big) = s_\Phi(f, \kappa). \tag{2.15}$$

Applying the "only if" part of Proposition 2 to the subpartition (t_0, t_{j-1}, t_j, t_n) of $(t_i)_{i=0}^n$, it follows that

$$\begin{aligned} v_\Phi\big(f, [a,b]\big) &= v_\Phi\big(f, [t_0, t_{j-1}]\big) + v_\Phi\big(f, [t_{j-1}, t_j]\big) + v_\Phi\big(f, [t_j, t_n]\big) \\ &= s_\Phi\big(f, (t_i)_{i=0}^{j-1}\big) + s_\Phi(f, \kappa) + s_\Phi\big(f, (t_i)_{i=j}^n\big), \end{aligned}$$

where the last equality holds by Corollary 3 applied twice and by (2.15). Therefore the merger partition $(t_i)_{i=0}^{j-1} \sqcup \kappa \sqcup (t_i)_{i=j}^n$ is maximizing for the Φ-variation of f. Also, since x is a partition point of κ, $x \in \mathcal{MPP}_\Phi(f, [a,b])$. This contradiction proves the second part of (2.14). $\square$

DEFINITION 3. Let $\Phi \in \mathcal{V}$, let $f \in \mathcal{PM}[a,b]$ for some $a, b \in \mathbb{R}$ with $a < b$, and let $t_a, t_b \in [a,b]^\pm$ be such that $a \leqslant t_a < t_b \leqslant b$. We say that t_a and t_b are *adjacent maximizing partition points* if there is a maximizing partition $\kappa = (t_i)_{i=0}^n$ for the Φ-variation of f such that $[t_a, t_b] = [t_{i-1}, t_i]$ for some i.

Lemma 2. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a,b]$ for some $a, b \in \mathbb{R}$ with $a < b$, and let t_a, t_b be adjacent maximizing partition points for the Φ-variation of f. Then*:

(i) *$v_\Phi(f, [t_a, t_b]) = \Phi(|f(t_b) - f(t_a)|)$, and*
(ii) *for each $x \in [t_a, t_b]^\pm$, $f(x)$ is between $f(t_a)$ and $f(t_b)$.*

Proof. Statement (i) is a particular case of Corollary 3. To prove (ii), let $x \in [t_a, t_b]^\pm$ and suppose that $f(t_a) \leqslant f(t_b)$. If $f(x) > f(t_b)$, then

$$\Phi\big(\big|f(t_b) - f(t_a)\big|\big) < \Phi\big(\big|f(x) - f(t_a)\big|\big) \leqslant v_\Phi\big(f, [t_a, t_b]\big).$$

This contradicts (i), and so $f(x) \leqslant f(t_b)$. If $f(x) < f(t_a)$, then the same contradiction follows by proving $f(t_a) \leqslant f(x)$. In the case $f(t_a) \geqslant f(t_b)$ the same argument completes the proof of (ii). $\square$

Lemma 3. *Let $\psi\colon [0,\infty) \to \mathbb{R}$ be a convex function, $d \geqslant 0$, and $D \in \mathbb{R}$. The implication*

$$\text{if } \psi(u+d) > \psi(u) + D, \text{ then } \psi(v+d) > \psi(v) + D \tag{2.16}$$

holds for any real numbers $0 \leqslant u \leqslant v$.

Proof. If either $d = 0$ or $u = v$, then (2.16) clearly holds. Let $d > 0$ and $0 \leqslant u < v$. It suffices to prove that

$$\psi(u+d) - \psi(u) \leqslant \psi(v+d) - \psi(v). \tag{2.17}$$

Since ψ is convex, we have

$$\frac{\psi(z) - \psi(x)}{z - x} \leqslant \frac{\psi(y) - \psi(x)}{y - x} \leqslant \frac{\psi(y) - \psi(z)}{y - z} \tag{2.18}$$

for any real numbers $0 < x < z < y$. First, taking $x = u$, $z = u + d$ and $y = v + d$, by the first inequality in (2.18) we have

$$\frac{\psi(u+d) - \psi(u)}{d} \leqslant \frac{\psi(v+d) - \psi(u)}{v - u + d}.$$

Second, taking $x = u$, $z = v$, and $y = v + d$, by the second inequality in (2.18) we have

$$\frac{\psi(v+d) - \psi(u)}{v - u + d} \leqslant \frac{\psi(v+d) - \psi(v)}{d}.$$

Combining the preceding two inequalities, (2.17) follows. The proof is complete. □

The next theorem is the main result of Section 2. The stated implication is also used in Section 3 in the equivalent form: if $x \notin \mathcal{MPP}_\Phi(f, [a', b'])$, then $x \notin \mathcal{MPP}_\Phi(f, [a, b])$. This means that if a point x is redundant in an interval $[a', b']$, then it remains redundant in a larger interval $[a, b]$.

Theorem 3. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a, b]$, and let $x \in [a', b'] \subset [a, b]$ with real numbers $a \leqslant a' < b' \leqslant b$. If $x \in \mathcal{MPP}_\Phi(f, [a, b])$, then $x \in \mathcal{MPP}_\Phi(f, [a', b'])$.*

Proof. Without loss of generality, we can assume that $[a', b'] \neq [a, b]$ and $a' < x < b'$. Assume that $x \in \mathcal{MPP}_\Phi(f, [a, b])$ and $x \notin \mathcal{MPP}_\Phi(f, [a', b'])$. It suffices to show that this assumption leads to a contradiction. By assumption there is $(t_i)_{i=0}^n \in \mathcal{MP}_\Phi(f, [a, b])$ such that $x = t_r$ for some $r \in \{1, \dots, n-1\}$. Let $(y_j)_{j=0}^m$ be any partition in $\mathcal{MP}_\Phi(f, [a', b'])$. By Lemma 1 applied to $[a', b']$ there exists $j \in \{1, \dots, m\}$ such that $y_{j-1} < x < y_j$ and $x \notin \mathcal{MPP}_\Phi(f, [y_{j-1}, y_j])$. Let l and k be, respectively, maximal and minimal $i \in \{1, \dots, n-1\}$ such that $t_i \in (y_{j-1}, y_j)$. Thus $k \leqslant r \leqslant l$, and

$$t_{k-1} \leqslant y_{j-1} < t_k \leqslant x = t_r \leqslant t_l < y_j \leqslant t_{l+1}.$$

The inequality (cf. (2.8))

$$v_\Phi\bigl(f, [y_{j-1}, y_j]\bigr) \geqslant v_\Phi\bigl(f, [y_{j-1}, t_k]\bigr) + v_\Phi\bigl(f, [t_k, t_l]\bigr) + v_\Phi\bigl(f, [t_l, y_j]\bigr) \tag{2.19}$$

holds by the definition of the Φ-variation. Suppose that the equality in (2.19) holds. Since f restricted to a subinterval remains piecewise monotone, by Corollary 1 there exist maximazing partitions κ_k and κ_l of $[y_{j-1}, t_k]$ and $[t_l, y_j]$, respectively, for the Φ-variation. If $k < l$, then by Corollary 3 $v_\Phi(f, [t_k, t_l]) = s_\Phi(f, (t_i)_{i=k}^l)$. Otherwise, $v_\Phi(f, [t_k, t_l]) = 0$. Therefore we have

$$v_\Phi\bigl(f, [y_{j-1}, y_j]\bigr) = s_\Phi(f, \kappa_k) + s_\Phi\bigl(f, (t_i)_{i=k}^l\bigr) + s_\Phi(f, \kappa_l).$$

Since $x = t_r$ and $(y_j)_{j=0}^m \in \mathcal{MP}_\Phi(f, [a', b'])$, by Corollary 3 it follows that $x \in \mathcal{MPP}_\Phi(f, [a', b'])$. This is a contradiction to the assumption, and so the equality in (2.19) is not possible. Therefore the strict inequality

$$v_\Phi\big(f, [y_{j-1}, y_j]\big) > v_\Phi\big(f, [y_{j-1}, t_k]\big) + v_\Phi\big(f, [t_k, t_l]\big) + v_\Phi\big(f, [t_l, y_j]\big) \tag{2.20}$$

must hold. Since y_{j-1} and y_j are adjacent maximizing partition points with respect to $(y_j)_{j=0}^m$, by Lemma 2(i) the left side of (2.20) is equal to $\Phi(|f(y_j) - f(y_{j-1})|)$. By the definition of the Φ-variation it then follows that

$$\Phi\big(\big|f(y_j) - f(y_{j-1})\big|\big) > \Phi\big(\big|f(t_k) - f(y_{j-1})\big|\big) + v_\Phi\big(f, [t_k, t_l]\big) + \Phi\big(\big|f(y_j) - f(t_l)\big|\big). \tag{2.21}$$

We next show that in the preceding inequality we can replace y_j by t_{l+1} and y_{j-1} by t_{k-1}. Without loss of generality, we can and do assume that $f(y_{j-1}) < f(y_j)$. Since (y_{j-1}, y_j), (t_{k-1}, t_k), and (t_l, t_{l+1}) are adjacent points with respect to the maximizing partition, using statement (ii) of Lemma 2 three-times it follows that

$$f(y_{j-1}) \leqslant f(t_l) \leqslant f(y_j) \leqslant f(t_{l+1}) \tag{2.22}$$

and

$$f(t_{k-1}) \leqslant f(y_{j-1}) \leqslant f(t_k) \leqslant f(t_{l+1}). \tag{2.23}$$

By (2.22) we have $d := f(t_l) - f(y_{j-1}) \geqslant 0$ and

$$v := f(t_{l+1}) - f(t_l) \geqslant f(y_j) - f(t_l) =: u \geqslant 0.$$

Letting $D := \Phi(|f(t_k) - f(y_{j-1})|) + v_\Phi(f, [t_k, t_l])$, by Lemma 3 and (2.21) we have the strict inequality

$$\Phi\big(\big|f(t_{l+1}) - f(y_{j-1})\big|\big) > \Phi\big(\big|f(t_k) - f(y_{j-1})\big|\big) + v_\Phi\big(f, [t_k, t_l]\big) + \Phi\big(\big|f(t_{l+1}) - f(t_l)\big|\big).$$

By similar arguments, in the preceding inequality, we replace y_{j-1} by t_{k-1}. By (2.23) we have $d := f(t_{l+1}) - f(t_k) \geqslant 0$ and

$$v := f(t_k) - f(t_{k-1}) \geqslant f(t_k) - f(y_{j-1}) =: u \geqslant 0.$$

Letting $D := v_\Phi(f, [t_k, t_l]) + \Phi(|f(t_{l+1}) - f(t_l)|)$, by Lemma 3 and (2.21) we obtain the strict inequality

$$\begin{aligned}\Phi\big(\big|f(t_{l+1}) - f(t_{k-1})\big|\big) &> \Phi\big(\big|f(t_k) - f(t_{k-1})\big|\big) + v_\Phi\big(f, [t_k, t_l]\big) + \Phi\big(\big|f(t_{l+1}) - f(t_l)\big|\big) \\ &= v_\Phi\big(f, [t_{k-1}, t_k]\big) + v_\Phi\big(f, [t_k, t_l]\big) + v_\Phi\big(f, [t_l, t_{l+1}]\big),\end{aligned}$$

where the preceding equality holds because (t_{k-1}, t_k) and (t_l, t_{l+1}) are adjacent points with respect to the maximizing partition $(t_i)_{i=0}^n$. This, together with Corollary 2, yields the inequality

$$\Phi\big(\big|f(t_{l+1}) - f(t_{k-1})\big|\big) > v_\Phi\big(f, [t_{k-1}, t_{l+1}]\big),$$

which contradicts the definition of the Φ-variation. Therefore the stated implication must hold. The proof of Theorem 3 is complete. □

Corollary 4. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a, b]$ for some $a, b \in \mathbb{R}$ with $a < b$, and let $\lambda = (x_i)_{i=0}^n$ be a partition of $[a, b]$. If*

$$v_\Phi\big(f, [a, b]\big) > \sum_{i=1}^n v_\Phi\big(f, [x_{i-1}, x_i]\big), \tag{2.24}$$

then at least one inner partition point of λ is redundant for the Φ-variation of f.

Proof. Suppose that (2.24) holds and each partition point of λ is maximizing for the Φ-variation of f. By Theorem 3, $x_i \in \mathcal{MPP}_\Phi(f, [x_{i-1}, b])$ for each $i \in \{1, \dots, n-1\}$. Then for each $i \in \{1, \dots, n-1\}$, (x_{i-1}, x_i, b) is a subpartition of some maximizing partition for the Φ-variation of f restricted to $[x_{i-1}, b]$, and so

$$v_\Phi\bigl(f, [x_{i-1}, b]\bigr) = v_\Phi\bigl(f, [x_{i-1}, x_i]\bigr) + v_\Phi\bigl(f, [x_i, b]\bigr)$$

by Proposition 2. Iterating the preceding equalities for $i \in \{1, \dots, n-1\}$, it follows that

$$v_\Phi\bigl(f, [a, b]\bigr) = v_\Phi\bigl(f, [a, x_1]\bigr) + v_\Phi\bigl(f, [x_1, b]\bigr) = \cdots = \sum_{i=1}^{n} v_\Phi\bigl(f, [x_{i-1}, x_i]\bigr),$$

which contradicts to hypothesis (2.24). Therefore at least one inner partition point of λ must be redundant for the Φ-variation of f whenever (2.24) holds. □

Let $\mathcal{K}$ be a nonempty set of subpartitions of a partition μ of $[a, b]$. Clearly, $\mathcal{K}$ contains at most finitely many elements. We say that a partition κ of $[a, b]$ is the *union partition of* $\mathcal{K}$ provided it contains all those and only those inner partition points that are inner partition points of at least one partition in $\mathcal{K}$.

Corollary 5. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a, b]$ for some $a, b \in \mathbb{R}$ with $a < b$, and let $\mu \in \mathcal{P}[a, b]$ be a minimal monotonicity partition for f. Let $\mathcal{K}_\mu$ be the set of subpartitions of μ which are maximizing partitions for the Φ-variation of f. Then the union partition of $\mathcal{K}_\mu$ is also a maximizing partition for the Φ-variation of f.*

Proof. The set $\mathcal{K}_\mu$ is nonempty by Theorem 1. Let $\lambda = (x_i)_{i=0}^n$ be the union partition of $\mathcal{K}_\mu$. Then each inner partition point of λ is maximizing, and so (2.9) holds by the preceding corollary. By the second part of Proposition 2 there is a maximizing partition $\kappa \in \mathcal{MP}_\Phi(f, [a, b])$ such that $\lambda \sqsubset \kappa \sqsubset \mu$. Therefore $\lambda = \kappa$ by the definition of λ, and the conclusion holds. □

Given a regulated function $f : [a, b] \to \mathbb{R}$, a point $x \in [a, b]^\pm$ is called a *global extremum* of f if $f(x)$ is either a largest or a smallest among all values of f. To achieve the supremum in the definition of Φ-variation, a partition need not contain all local extremum points. But the global extremum points must be partition points of some maximizing partition as the following shows.

Proposition 3. *Let $\Phi \in \mathcal{CV}$, and let $f \in \mathcal{PM}[a, b]$ for some $a, b \in \mathbb{R}$ with $a < b$. Each global extremum of f is a partition point of a maximizing partition for the Φ-variation of f.*

Proof. Let $x \in [a, b]^\pm$ be a global extremum of f, and let $(t_i)_{i=0}^n$ be a maximizing partition for the Φ-variation of f. There is $j \in \{1, \dots, n\}$ such that $x \in [t_{j-1}, t_j]^\pm$. We can suppose that $t_{j-1} < x < t_j$ since otherwise the proof is complete. Since x is a global extremum, we have

$$\bigl|f(t_j) - f(t_{j-1})\bigr| \leqslant \max\bigl\{\bigl|f(x) - f(t_{j-1})\bigr|, \bigl|f(t_j) - f(x)\bigr|\bigr\}.$$

Therefore

$$\Phi\bigl(\bigl|f(t_j) - f(t_{j-1})\bigr|\bigr) \leqslant v_\Phi\bigl(f, [t_{j-1}, x]\bigr) + v_\Phi\bigl(f, [x, t_j]\bigr).$$

Since (t_{j-1}, t_j) is a pair of adjacent maximizing partition points, the left side of the preceding inequality is equal to $v_\Phi(f, [t_{j-1}, t_j])$. Since the reverse inequality always holds, in fact, we have the equality

$$v_\Phi\bigl(f, [t_{j-1}, t_j]\bigr) = v_\Phi\bigl(f, [t_{j-1}, x]\bigr) + v_\Phi\bigl(f, [x, t_j]\bigr). \tag{2.25}$$

On the other hand, by the "only if" part of Proposition 2 applied to the subpartition $\lambda = (a, t_{j-1}, t_j, b)$ of $(t_i)_{i=0}^n$, and then by (2.25) it follows that

$$v_\Phi\bigl(f, [a, b]\bigr) = v_\Phi\bigl(f, [a, t_{j-1}]\bigr) + v_\Phi\bigl(f, [t_{j-1}, x]\bigr) + v_\Phi\bigl(f, [x, t_j]\bigr) + v_\Phi\bigl(f, [t_j, b]\bigr).$$

By the "if" part of Proposition 2 applied to the partition $\lambda = (a, t_{j-1}, x, t_j, b)$ there is a maximizing partition κ for the Φ-variation of f such that $\lambda \sqsubset \kappa$, and so x is a partition point of κ. The proof is complete. □

Among the hypotheses of the following statement, the number $K(f)$ defined by (2.5) is used.

Lemma 4. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a,b]$ for some $a, b \in \mathbb{R}$ with $a < b$, and let $K(f)$ be even. Then $[a,b]^{\pm} \setminus \{a,b\}$ contains a partition point of a maximizing partition for the Φ-variation of f.*

Proof. Let $(t_i)_{i=0}^{K(f)} \in \Xi(f)$. Since $K(f)$ is even integer, $K(f) > 1$. Then by Proposition 1 either its statement (i) holds, or its statement (ii) holds. Suppose that (i) holds. Thus $f(t_{n-1}) < f(t_n)$, and so b is not a global minimum. Also, since $f(t_0) > f(t_1)$, a is not a global minimum. Since each piecewise monotone function has a global minimum x, by Proposition 3 x must be a partition point of a maximizing partition for the Φ-variation, which is different from the endpoints a and b. If statement (ii) holds, then the symmetric argument applied to a global maximum yields the same conclusion. □

Lemma 5. *Let $\Phi \in \mathcal{CV}$, let $f \in \mathcal{PM}[a,b]$, and let $(t_i)_{i=0}^n$ be a partition of $[a,b]$. Suppose that*

$$(t_i)_{i=0}^{n-1} \in \mathcal{MP}_\Phi\big(f, [a, t_{n-1}]\big) \quad \textit{and} \quad (t_i)_{i=1}^{n} \in \mathcal{MP}_\Phi\big(f, [t_1, b]\big). \tag{2.26}$$

Then either

$$\forall i \in \{1, \dots, n-1\}: \quad t_i \in \mathcal{MPP}_\Phi\big(f, [a,b]\big), \tag{2.27}$$

or

$$\forall i \in \{1, \dots, n-1\}: \quad t_i \notin \mathcal{MPP}_\Phi(f, [a,b]). \tag{2.28}$$

Proof. Assume that (2.28) does not hold. We will prove that (2.27) holds. By the assumption we have

$$\exists j \in \{1, \dots, n-1\}: \quad t_j \in \mathcal{MPP}_\Phi\big(f, [a,b]\big).$$

This means that there exists $\kappa \in \mathcal{MP}_\Phi(f, [a,b])$ such that $t_j \in \kappa$. Applying the "only if" part of Proposition 2 to $\lambda = (a, t_j, b)$, we have that

$$v_\Phi\big(f, [a,b]\big) = v_\Phi\big(f, [a, t_j]\big) + v_\Phi\big(f, [t_j, b]\big).$$

By hypothesis (2.26) and by Corollary 3 we have

$$v_\Phi\big(f, [a, t_j]\big) = s_\Phi\big(f, (t_i)_{i=0}^{j}\big) \quad \text{and} \quad v_\Phi\big(f, [t_j, b]\big) = s_\Phi\big(f, (t_i)_{i=j}^{n}\big).$$

It then follows that

$$v_\Phi\big(f, [a,b]\big) = s_\Phi\big(f, (t_i)_{i=0}^{j}\big) + s_\Phi\big(f, (t_i)_{i=j}^{n}\big) = s_\Phi\big(f, (t_i)_{i=0}^{n}\big).$$

Therefore $(t_i)_{i=0}^n \in \mathcal{MP}_\Phi(f, [a,b])$, and so statement (2.27) holds. □

3 Construction of algorithm

In this section, we present an algorithm for calculating the p-variation of a piecewise constant function for any $0 < p < \infty$. Nonetheless, this algorithm can be used to calculate the p-variation of an arbitrary piecewise monotone function. This procedure is called *pvar*. It is already implemented in the R package *pvar* (see [14]) and is publicly available on CRAN.[1]

First, we give some introductory notes about the procedure. Second, a basic scheme of the algorithm is presented in Section 3.2. Third, each step of the algorithm is discussed with more details in Section 3.3.

[1] `http://cran.r-project.org/web/packages/pvar/index.html`

3.1 Introductory notes

We show that given a piecewise monotone function f on an interval $[a,b]$, without changing its p-variation, we can replace f by a right-continuous piecewise constant function on an interval $[0,n]$ with values $f(t_i)$, $i \in \{0,1,\dots,n\}$, where $(t_i)_{i=0}^n \in \Xi(f)$ and $n = K(f)$. In other words, we show that p-variation of a piecewise monotone function f depends only on a tuple $(f(t_0), f(t_1), \dots, f(t_n))$ of values on a minimal monotonicity partition $(t_i)_{i=0}^n$ of $[a,b]^{\pm}$.

To show that p-variation of an arbitrary piecewise monotone function can be computed using a suitable piecewise constant function, we use a few more facts from Qian work [12].

Proposition 4. (*See* [12, *Prop.* 3.3].) *Let $f \in \mathcal{PM}[a,b]$ for some $a,b \in \mathbb{R}$ with $a < b$. If (t_j) and (s_j) are partitions in $\Xi(f)$, then $f(t_j) - f(t_{j-1}) = f(s_j) - f(s_{j-1})$ for each $j \in \{1,\dots,K(f)\}$.*

With a piecewise monotone function f on $[a,b]$, we can associate the tuple $(\alpha_j(f))_{j=1}^{K(f)}$ of real numbers defined by $\alpha_j(f) := f(t_j) - f(t_{j-1})$, $j \in \{1,\dots,K(f)\}$, for some (any) $(t_j) \in \Xi(f)$. If f is not a constant, then the numbers $\alpha_j(f) \neq 0$ have alternating signs.

Definition 4. (See [12, Def. 3.4].) If f and g are piecewise monotone functions, possibly defined on different intervals, such that $K := K(f) = K(g)$ and $\alpha_j(f) = \alpha_j(g)$ for $j \in \{1,2,\dots,K\}$, then we say that f and g are $\mathcal{PM}$-equivalent and write $f \overset{\mathcal{PM}}{=} g$.

For example, given a piecewise monotone function f and a real number c, f and $c + f$ are $\mathcal{PM}$-equivalent. Using Theorems 1 and 2, we obtain the next fact.

Corollary 6. *Let $p \in (0,\infty)$, let f and g be piecewise monotone functions, possibly defined on different intervals. If $f \overset{\mathcal{PM}}{=} g$ or $f \overset{\mathcal{PM}}{=} -g$, then $v_p(f) = v_p(g)$.*

Now we are prepared to formulate the main concept for this section.

Definition 5. Any tuple $X = (X_i)_{i=0}^n = (X_0, X_1 \dots, X_n)$ of real numbers is called a *sample* with a *sample size* n denoted by $|X|$. The function $G_X : [0,n] \to \mathbb{R}$ with values

$$G_X(t) := X_{\lfloor t \rfloor}, \quad t \in [0,n],$$

is called the (associated) *sample function*, where $\lfloor t \rfloor$ is the value of the floor function at point t. Let $p \in (0,\infty)$. The p-variation of a sample X is defined to be the p-variation of the sample function G_X and is denoted by

$$v_p(X) := v_p\bigl(G_X, [0,n]\bigr).$$

Let f be a piecewise monotone function defined on $[a,b]$, and let $X(f)$ be a sample $(f(t_0), f(t_1), \dots, f(t_n))$ with sample size $|X(f)| = n = K(f)$, where partition $(t_i)_{i=0}^n \in \Xi(f)$. By definition of the associated sample function $G_{X(f)}$ we have $f \overset{\mathcal{PM}}{=} G_{X(f)}$, and so $v_p(f) = v_p(X(f))$, provided that $p \in (0,\infty)$. From now on, we consider a sample X and the associated sample function G_X.

Let X be a sample with a sample size $|X| = n$. By definition (2.3), G_X is constant on each open interval $(i-1,i)$, $i \in \{1,\dots,n\}$, and so it is piecewise constant function with minimal monotonicity partition $\eta = (0,1,\dots,n)$ of $[0,n]$. Clearly, G_X is right continuous.

Let $p \in (0,1]$. By Theorem 2, since the associated sample function G_X is right-continuous, we have

$$v_p(X) = s_p(G_X, \eta) = \sum_{i=1}^n |X_i - X_{i-1}|^p.$$

In this case, η is a maximizing partition for the p-variation of X. Let $p \in (1, \infty)$. By Theorem 1 the supremum of the p-variation of X is achievable on a subpartition of partition $\eta = (0, 1, \dots, n)$, and so

$$v_p(X) = \max\left\{ \sum_{i=1}^{k} |X_{j_i} - X_{j_{i-1}}|^p \colon (j_0, j_1, \dots, j_k) \sqsubset \eta \right\}.$$

The main task when computing the p-variation of X is to find a maximizing partition (Definition 1), which is a subpartition $(j_0, j_1, \dots, j_k)$ of η in this case. Further on, we identify a subpartition $(j_0, j_1, \dots, j_k)$ of η with the subsample $S = (S_0, S_1, \dots, S_k) := (X_{j_0}, X_{j_1}, \dots, X_{j_k})$ of $X$and apply the corresponding notation to subsamples.

Let $p \in (1, \infty)$, and let $X = (X_i)_{i=0}^n$ be a sample. An element X_i of X is a maximizing partition point for the p-variation of X if it is an element of some maximizing subpartition of X for the p-variation of X (cf. Definition 2). If an element X_i of X is not a maximizing partition point for the p-variation of X, then it is called a *redundant point for the* p*-variation of* X, or just a *redundant point* when the sample X is clear from the context.

Redundant points of the sample X can be removed from X without changing the value of the p-variation of X. The main task of the algorithm is to identify all redundant points in a sample X. To this aim, we use Theorem 3 and other facts from Section 2.2. By this theorem it suffices to find a redundant point of a small subpartition of X. Once such a point is found and removed from X, the same procedure is repeated until the remaining sample S contains only a maximal maximizing partition for the p-variation of X.

3.2 The main function of the *pvar* procedure

The main function describes a general scheme of calculating the p-variation of a given sample. Details of each step are not described here, since they are discussed in the next subsection. The main steps in *pvar* procedure are as follows.

Procedure *pvar*. Input: sample X, scalar p.

1. *Initialization and checking for exceptions.* At the beginning, a whole sample is considered, so $S \leftarrow X$. First, we verify particular cases. For example, if $|X| = 0$, then X has only one element. In this case, the p-variation is equal to 0, and so the procedure is finished. Also, if $0 < p \leqslant 1$, then all elements of X are maximizing partition points, and again the procedure is finished by giving the value of p-variation. Further on, we assume that $|X| > 0$ and $p > 1$.
2. *Finding minimal monotonicity partition.* To this aim, we find all points of strict local extrema of the sample function. According to Theorem 1, all points that are not strict local extremum points can be excluded from further consideration.
3. *Checking small subsamples.* Every small subsample is checked for redundant points. These points are also redundant points for the initial sample by Theorem 2. So, if X_j appears to be redundant in any small subsample, then it is removed. More details are given in Section 3.3.2.
4. *Merging small subsamples.* Let $S_{0,i}$ and $S_{i,n}$ be maximizing partitions for the p-variation over intervals $[0, i]$ and $[i, n]$, respectively. The union sample $S_{0,i} \sqcup S_{i,n}$ is used to find a maximizing partition $S_{0,n}$ over the interval $[0, n]$. Finding $S_{0,n}$ from $S_{0,i}$ and $S_{i,n}$ is called *merging*. Merging is repeated for all small maximizing partitions until a final maximizing partition is found. In Section 3.3.3, the whole step is described in detail.
5. *Calculating the value of* p*-variation.* When the maximizing partition $(S_i)_{i=0}^n$ is obtained, the value of p-variation is calculated using the p-variation sum

$$v_p = \sum_{i=1}^{n} |S_{i-1} - S_i|^p$$

The corresponding pseudo-code of the main function looks as follows.

Algorithm 1. The main function of *pvar* procedure that calculates the p-variation of a sample

Input: sample X, scalar p.
Require: $p > 0$.
Output: the value of p-variation of X

function PVAR(X, p)
    $S \leftarrow X$;
    if $0 < p \leqslant 1$ **then**
        value $\leftarrow \sum |S_{i-1} - S_i|^p$;
        return value;
    end if
    if $|S| = 0$ **then**
        value $\leftarrow 0$;
        return value;
    end if
    $S \leftarrow FindCorners(S)$;
    $M \leftarrow 3$;
    $S \leftarrow CheckSubSamples(S, p, M)$;
    $S \leftarrow MergeAll(S, p, M + 1)$;
    value $\leftarrow \sum |S_{i-1} - S_i|^p$;
    return value;
end function

The function *FindCorners* returns the minimal monotonicity partition of a sample (see Section 3.3.1). The function *CheckSubSamples* checks all small subsamples with size up to some M and removes all the points that appeared to be redundant (see Section 3.3.2). Function *MergeAll* merges all small subsamples until a final maximizing partition is obtained (see Section 3.3.3).

3.3 Detailed explanation

3.3.1 Finding minimal monotonicity partition

By Theorem 1 a maximizing partition of the p-variation of a sample is a subpartition of a minimal monotonicity partition. By Proposition 1 each inner partition point of a minimal monotonicity partition is a strict local extremum of a sample function. From now on, a partition point is called a *corner* if it is either an endpoint or a strict local extremum. To find all corners in a given a sample $S = (S_0, S_1, \ldots, S_n)$, we look for a change of sign of increments when passing from $S_i - S_{i-l}$ for some l to $S_{i+1} - S_i$. The endpoints S_0 and S_n are the corners by definition. The following pseudo-code removes from the set $\{S_1, \ldots, S_{n-1}\}$ all elements that are not corners.

Algorithm 2. The function *FindCorners* that returns the subset of S, which contains only the corners of S

Input: sample S
Output: sample $corners$ that is the subset of S, which contains only the corners

function FINDCORNERS(S)
    $corners \leftarrow S_0$;
    $direction \leftarrow 0$;
    $n \leftarrow |S|$;
    for $i \leftarrow 1$ **to** $n - 1$ **do**
        if $S_{i-1} < S_i$ **then**
            if $direction < 0$ **then**
                append $corners$ with element S_{i-1};
            end if
            $direction \leftarrow 1$;
        end if

```
        if S_{i-1} > S_i then
            if direction > 0 then
                append corners with element S_{i-1};
            end if
            direction ← −1 ;
        end if
    end for
    append corners with element S_n;
    return corners;
end function
```

This procedure requires very little computing time since it does not make any cross check and has only one loop. The function *FindCorners* in *pvar* package is under the name *ChangePoints*.

3.3.2 Checking small subsamples

Let $S = (S_0, S_1, \dots, S_n)$ be a minimal monotonicity partition, that is, each partition point of S is a corner. Let $m \in \{2, \dots, n-1\}$ and $k \in \{0, \dots, n-m\}$. If

$$\sum_{i=k+1}^{k+m} |S_i - S_{i+1}|^p < |S_{k+m} - S_k|^p, \tag{3.1}$$

then some of the points in $(S_k, \dots, S_{k+m})$ must be redundant for the p-variation of S by Corollary 4. Lemmas 4 and 5 are used to identify redundant points step by step starting with small m.

1. Let $m = 2$. A subsample (S_k, S_{k+1}, S_{k+2}) of S has exactly one inner partition point, which must be a maximizing partition point by Lemma 4. Therefore (3.1) is true for no values of $k \in \{0, \dots, n-2\}$, and so each partition (S_k, S_{k+1}, S_{k+2}) is maximizing.
2. Let $m = 3$. If (3.1) holds for a subsample of size $m = 3$, then at least one inner partition point must be redundant. If so, then both inner partition points must be redundant by Lemma 5, which applies since all subsamples (S_k, S_{k+1}, S_{k+2}) of size $m = 2$ are maximizing by the preceding step. Thus both inner points should be removed. Property (3.1) is verified successively for each $k = 0, \dots, n-3$. If at least one redundant partition point is found, then property (3.1) is verified once again for each $k = 0, \dots, n-3$. Otherwise, all remaining partitions of size $m = 3$ are maximizing, and we proceed to the next step.
3. Let $m = 4$. At least one inner partition point of a subsample $(S_k, \dots, S_{k+4})$ must be maximizing by Lemma 4 since its size is even. Applying Lemma 5 as before, we can conclude that all inner partition points are maximizing. Note that this argument holds whenever the sample size is even and all redundant partition points are removed from each subsample with odd sample size in the preceding step. Therefore we skip the steps corresponding to subsamples with even size.
4. Let $m = 5$, let all subsamples of size $m = 3$ be already verified, and let $(S_k, \dots, S_{k+5})$ be a subsample with $k \in \{0, \dots, m-5\}$. If (3.1) holds for this subsample, then all inner points S_{k+1}, S_{k+2}, S_{k+3}, and S_{k+4} are redundant by Lemma 5 and are removed. Before repeating this procedure, we need to begin with subsamples of size $m = 3$, which is needed to use Lemma 5.
5. To verify subsamples of size $m = 7, 9, 11, \dots$, we have to begin with the case $m = 3$ and increase m by 2.

We can continue this checking until m reaches some threshold M. The pseudo-code of this procedure is given next.

Algorithm 3. Procedure *CheckSubSamples*, which ensures that any sequential subsample of length M is a maximizing partition in the corresponding interval

Input: sample S, scalar p, scalar M
Require: S must contain only the corners
Output: modified sample S such that any sequential subsample of length M is a maximizing partition in corresponding interval

function CHECKSUBSAMPLES(S, p, M)
  $tempS \leftarrow NULL$;
  $m \leftarrow 0$;
  while $tempS \neq S$ **and** $m < M$ **do**
    $tempS \leftarrow S$;
    $m \leftarrow 3$;
    check all subsamples of S with length m ;
    update S by dropping all redundant points ;
    while $tempS = S$ **and** $m < M$ **do**
      $m \leftarrow m + 2$;
      check all subsamples of S with length m ;
      update S by dropping all redundant points ;
    end while
  end while
  return S;
end function

By taking the threshold M sufficiently large this procedure can be used to calculate the p-variation of the original sample. But this strategy may not be effective in the case we need to repeat checking of subsamples of size $m = 3$. It is reasonable to use this procedure only for small m and then move to the next step of merging small subsamples.

3.3.3 Merging subsamples

Here we describe the code of the final step of computation of the p-variation of a sample. At this step, adjacent pairs of small maximizing subsamples are combined consecutively into a maximizing partition for the p-variation of the original sample. Given two subsamples that are maximizing partitions of two adjacent intervals $[a, w]$ and $[w, b]$, an operation producing a maximizing partition of the interval $[a, b]$ is called *merging*. This procedure has two stages to be discussed separately.

Merging two subsamples. The first stage describes merging of two adjacent maximizing subsamples. Suppose we are given two subsamples $(S_i)_{i=a}^{w}$ and $(S_i)_{i=w}^{b}$ that are maximizing partitions of the intervals $[a, w]$ and $[w, b]$, respectively. The point w is a maximizing point in each of them because it is an endpoint. However, w may not be a maximizing point in the union subsample $(S_i)_{i=a}^{b}$. By Theorem 3 a maximizing partition of the interval $[a, b]$ is some subsample of $(S_i)_{i=a}^{b}$.

To find a maximizing partition of $[a, b]$, it suffices to investigate values of the p-variation sums

$$s_p(S, \kappa_{i,j}) = \sum_{k=a+1}^{i} |S_k - S_{k-1}|^p + |S_j - S_i|^p + \sum_{k=j+1}^{b} |S_k - S_{k-1}|^p$$

over partitions $\kappa_{i,j} = (a, \dots, i, j, \dots, b)$ with i and j varying over $[a, w-1] := \{a, \dots, w-1\}$ and $[w, b] := \{w, \dots, b\}$, respectively. Note that

$$s_p(S, \kappa_{i,j}) = s_p(S, \kappa) + |S_j - S_i|^p - \sum_{k=i+1}^{j} |S_k - S_{k-1}|^p,$$

where $\kappa = (a, a+1, \ldots, b-1, b)$. Let

$$(t, r) := \underset{(i,j)\in[a,w-1]\times[w,b]}{\arg\max} \left\{ |S_j - S_i|^p - \sum_{k=i+1}^{j} |S_k - S_{k-1}|^p \right\}. \tag{3.2}$$

If $(t, r) = (w-1, w)$, then $(S_i)_{i=a}^{b}$ is a maximizing subpartition, and we do not make changes. Otherwise, we modify the calculation using Lemma 2: if t and r are adjacent maximizing partition points, then $(t, r) \in T$, where

$$T := \left\{ (i, j) \in [a, w-1] \times [w+1, b] \colon\ S_i \geqslant S_k \geqslant S_j \text{ or } S_i \leqslant S_k \leqslant S_j \ \ \forall k \in [i, j] \right\}.$$

Then (3.2) is replaced by

$$(t, r) := \underset{(i,j)\in T}{\arg\max} \left\{ |S_j - S_i|^p - \sum_{k=i+1}^{j} |S_k - S_{k-1}|^p \right\}.$$

Provided that a pair (t, r) of adjacent maximizing points is found, all the inner points $[t+1, r-1]$ are redundant and are removed. The remaining points comprise a maximizing partition.

Algorithm 4. Procedure *MergeSupremeSamples*, which merges two maximizing samples

Input: sample A, sample B, scalar p
Require: samples A and B must be maximizing partitions in corresponding intervals.
Output: sample S, which is the maximizing partition of joined interval

function MERGESUPREMESAMPLES(A, B, p)
    $S \leftarrow A \cup B$;
    $a \leftarrow 0$;
    $b \leftarrow |S|$;
    $w \leftarrow |A|$;
  // Find the T list:
    initiate T as empty list ;
    $cummax \leftarrow cummin \leftarrow S_w$;
    for $i \leftarrow w-1$ **to** a **do**
        if $S_i > cummax$ **then**
            $cummax \leftarrow S_i$;
            append T with element i ;
        end if
        if $S_i < cummin$ **then**
            $cummin \leftarrow S_i$;
            append T with element i ;
        end if
    end for
  // Find the R list:
    initiate R as empty list ;
    $cummax \leftarrow cummin \leftarrow S_w$;
    for $i \leftarrow w+1$ **to** b **do**
        if $S_i > cummax$ **then**
            $cummax \leftarrow S_i$;
            append R with element i ;
        end if
        if $X_i < cummin$ **then**
            $cummin \leftarrow S_i$;
            append R with element i ;
        end if
    end for

// Check all pairs in T x R:
$maxbalance \leftarrow 0$;
for all $r \in R$ **do**
  for all $t \in T$ **do**
    $balance \leftarrow |S_t - S_r|^p - \sum_{i=t+1}^{r} |S_{i-1} - S_i|^p$;
    if $balance > maxbalance$ **then**
      $maxT \leftarrow t$;
      $maxR \leftarrow r$;
      $maxbalance \leftarrow balance$;
    end if
  end for
end for
if $maxbalance > 0$ **then**
  remove points $maxT + 1, \dots, maxR - 1$ from sample S ;
end if
return S;
end function

In *pvar* package the merge operation can be achieved with *AddPvar* function.

Merging all subsamples. The second stage describes consecutive merging of pairs of adjacent maximizing partitions until a maximizing partition of the original sample is obtained. The pseudo-code of this procedure is given next.

Algorithm 5. Procedure *MergeAll*, which merges all subsamples of the size M into final maximizing sample of X.

Input: sample S; scalar p; scalar M.
Require: any subsample of S of length M must be a maximizing partition
Output: modified sample S that corresponds to maximizing portion of the whole sample
function MERGEALL(X, p, M)
  // construct the set $R_i, i = 1, \dots,$ which contain maximizing subsamples
  for $i \leftarrow 1$ **to** $\lceil \frac{|S|}{M} \rceil$ **do**
    $R_i \leftarrow \{S_j\}_{j=(i-1)M}^{\min(iM,|S|))}$;
  end for

  // apply *MergeSupremeSamples* to pairs of R elements until all intervals are merged
  while $|R| > 1$ **do**
    $i \leftarrow 1$;
    while $i < |R|$ **do**
      $R_i \leftarrow MergeSupremeSamples(R_i, R_{i+1}, p)$;
      $i \leftarrow i + 2$;
    end while
    delete all elements form R that have even index ;
  end while
  $S \leftarrow R_1$;
  return S;
end function

4 Conclusion

An algorithm of computation of the p-variation of a piecewise monotone function is given. The algorithm is based on properties of partitions attaining the supremum value of the p-variation proved in the first part of the paper. The algorithm is already available in R package under the name *pvar*. The cores of functions are written in C++ language. The algorithm is sufficiently fast and computes the p-variation of really complicated functions.

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